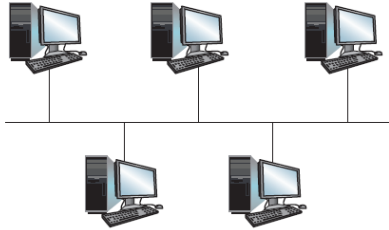
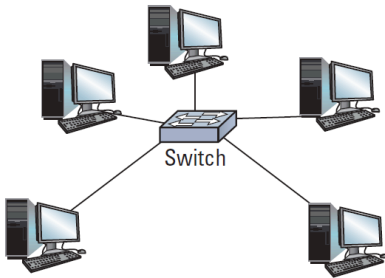


Network Classifications

- **Scope**
 - Personal area network (PAN)
 - Local area network (LAN)
 - Metropolitan area (MAN)
 - Wide area network (WAN)
- **Owner Ship**
 - Open Network e.g. Global Internet
 - Closed/Proprietary(專有) Network
- **Network Topology**: The pattern in which the machines are connected.
 - Bus: The machines are all connected to a common communication line. e.g. Ethernet

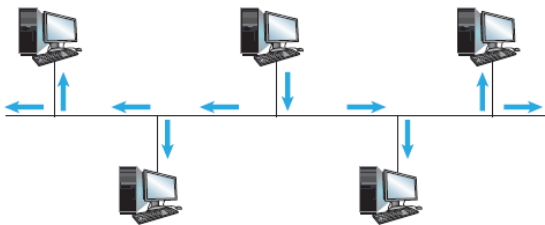


- Star: A large central computer serving many users. e.g. Wireless networks with central Access Point(AP)

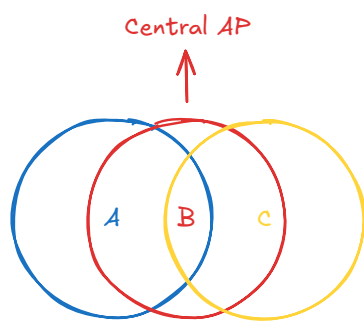


Protocols

- **Rules by which activities are conducted**
- **Carrier Sense Multiple Access/Collision Detection (CSMA/CD)**
 - Used in Ethernet.
 - Each message be broadcast to all the machines on the bus
 - To **transmit a message**, a machine waits until **the bus is silent** and monitors the bus in the same time.



- **Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA)**
 - Used in WIFI.
 - Cause of the **collision**
 - A machine's signal drowns out that of the other machine.
 - **Hidden terminal problem**: Each machine **can't detect other machine's signal** because it blocked by **distance or obstacle**.



A and C both of them can't detect their signals each other.

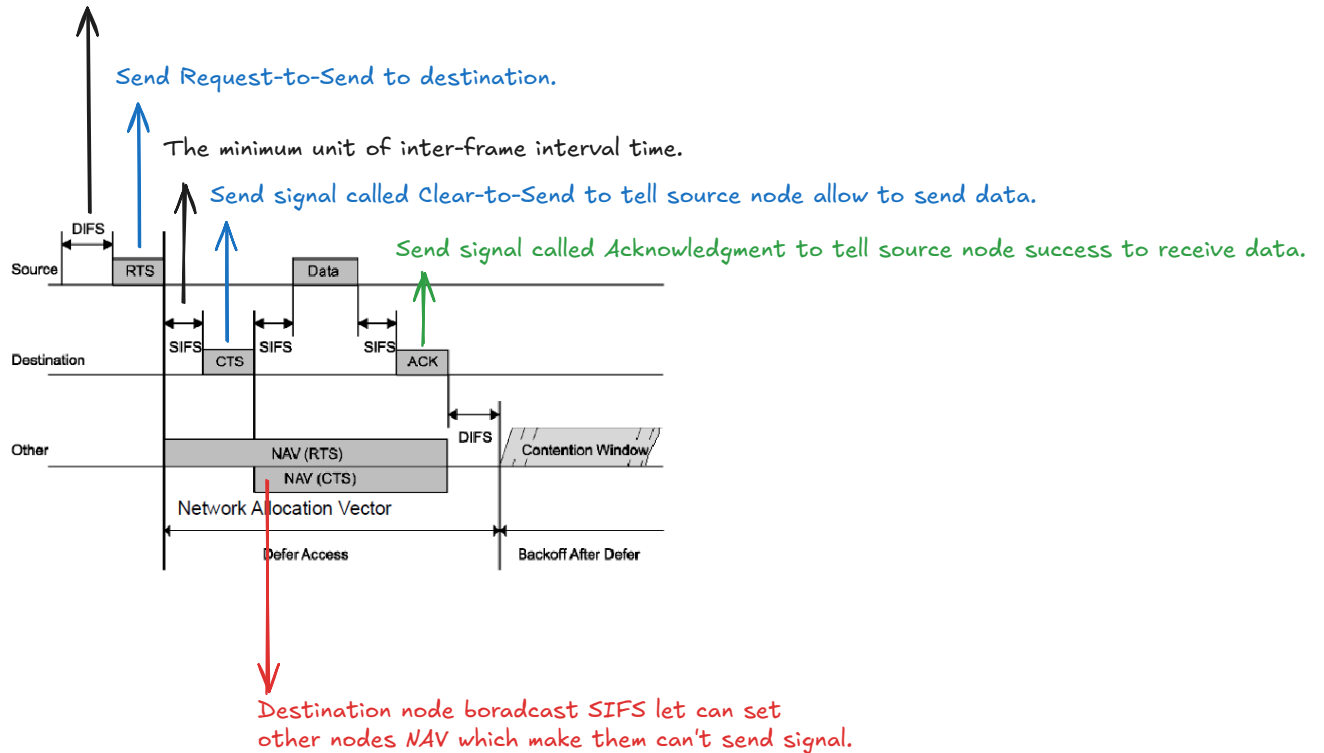
If A and C send signal to B in the same time

Collision

- Solve hidden terminal problem

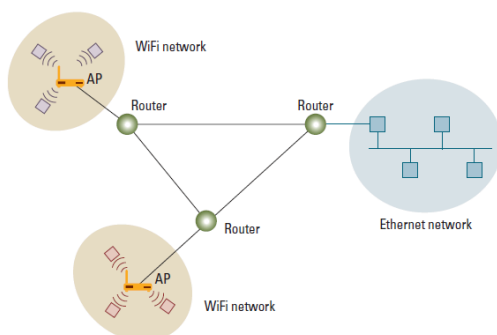
- Request-to-send/Clear-to-send (RTS/CTS) handshaking

Wait for a long time interval to ensure the channel is silent.



Connecting Network

- Repeater**: Extends a network. **Receives all signals** from a network and sends to the another network.
- Bridge**: Connects two **compatible** networks. **Checks eash signal's destination** and sends it to the destination.
- Switch**: Connects several **compatible** networks.
- Router**: Connects **incompatible** networks, builds a network of networks called **internet**.
 - Gateway: The terminal of network connecting to internet.
 - Forwarding table: Records the path which connects to other networks.

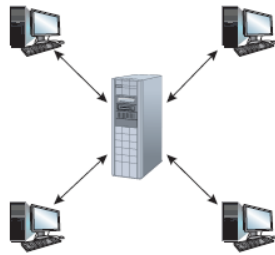


Inter-process Communication

- The process of coordinating each computer's activities in the network.

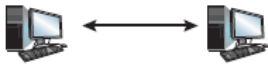
- **Client/Server Model**

- **One server corresponds to many clients.**
- Server must execute continuously.
- Client sends request to server.



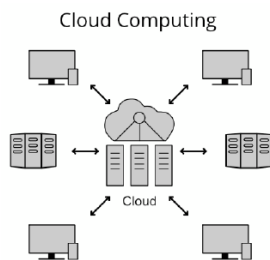
- **Peer to peer(P2P) Model:**

- Each process can provide and receive service each other.
- Execute on a temporary basis.



Distributed Systems

- Systems with parts that run on different computers.
- **Cluster Computing:** Many computers work closely together to provide computation or services.
- **Grid Computing:** Computers in different places work together for a task.
- **Cloud Computing:** All computers can use the shareable data in the pool on the network. e.g. Google Drive



Intranet

- A private, secure network for a single organization to share information among employees.

Internet

- **The Internet:** An internet that spans the world.
 - Original goal: Develop a worldwide connecting network and can't be affected by local disaster.
 - Today: A commercial undertaking that links a worldwide involving millions of computers.

Internet Architecture

- **Internet Service Providers (ISPs)**
 - **Tier1 ISPs:** high-speed, high-capacity, **international** WANs.
 - **Tier2 ISPs:** More regional.
 - Tier1 and Tier2 are Internet's Infrastructure and networks of routers.
 - **Tier3/Access ISP:** Provides connectivity to the Internet.
 - Hot spot (wireless)
 - Telephone lines
 - Cable/Satellite systems DSL
 - Fiber optics

Internet Addressing

- **IP address:** The address that lets the Network know each computer.

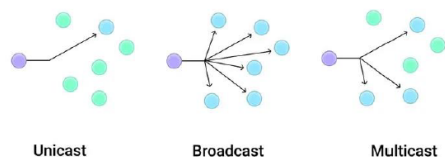
- Uses pattern of **32(IPv4) or 128(IPv6) bits** often represented in **dotted decimal notation**.

IPv4 Address Classes and Ranges						
Address Class	Type	Range	Default Subnet Mask	Number of Networks	No of Hosts Per Network	Use
A	Public	1.0.0.0 to 127.0.0.0	255.0.0.0	126	16,777,214	Governments and Large Number of Hosts
	Private	10.0.0.0 to 10.255.255.255				
B	Public	128.0.0.0 to 191.255.255.255	255.255.0.0	16,382	65,534	Medium Companies
	Private	172.16.0.0 to 172.31.255.255				
C	Public	192.0.0.0 to 223.255.255.255	255.255.255.0	2,097,150	254	Small Companies and LANs
	Private	192.168.0.0 to 192.168.255.255				
D	N/A	224.0.0.0 to 239.255.255.255	Not Applicable	N/A	N/A	Reserved for Multicasting
E	N/A	240.0.0.0 to 254.255.255.255	Not Applicable	N/A	N/A	Experimental
Special	Special	127.0.0.1 to 127.255.255.255	N/A	N/A	N/A	Loopback Testing
Note: - Addresses 127.0.0.1 to 127.255.255.255 cannot be used and are reserved for loopback testing - APIPA address range is 169.254.0.1 to 169.254.255.254 and has 65, 534 usable IP addresses, with the subnet mask of 255.255.0.0.						

- **MAC address:** The address which is assigned by the manufacturer and used to **identify devices on a local network**.
 - Consists of 12-character alphanumeric code. `00:0A:95:9D:68:16`
 - The **first six characters** used to **identify the manufacture**. `00:0A:`
- **Mnemonic address:** Easy to remember IP address.
 - Domain name: The name that identifies the **organization associated with an IP address**.
 - Expressed in several layers, separated by dot. e.g. `google.com`
 - Top-Level Domains(TLDs): The **rightmost part** of the domain name.
 - Indicates the domain's **feature**. e.g. `.com`(commercial), `.tw`(Taiwan), etc
- **Domain name system (DNS):** Turns the domain name into the numeric IP.
 - DNS lookup: The process of using DNS to perform a translation.
 - Name server: The place which executes DNS lookup.
- **Internet Corporation for Assigned Names & Numbers (ICANN)**
 - Allocates IP addresses to ISPs who then assign those addresses within their regions.
 - Oversees the registration of domains and domain names.

Internet Application

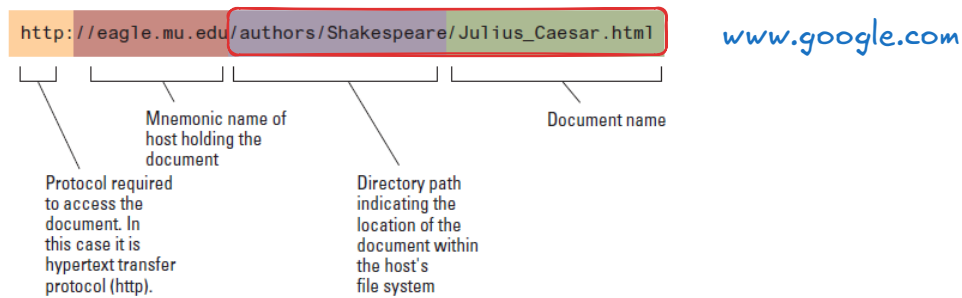
- **Network News Transfer Protocol (NNTP)**
- **File Transfer Protocol FTP (port 21)**
- **Telnet and SSH (port 22)**
- **Hypertext Transfer Protocol (HTTP) (port 80)**
- **Hypertext Transfer Protocol Secure (HTTPS) (port 443)**
- **Internet Control Message Protocol (ICMP) used to send error messages and operational information**
- **Electronic Mail**
 - Domain mail server: Collects incoming mail and transmits outgoing mail.
 - Mail server uses POP3(Post Office Protocol version 3) or IMAP(Internet Message Access Protocol) to send mails to the clients.
- ****Voice over IP(VoIP)**
- **Internet Multimedia Streaming**
 - N-unicast
 - Multicast
 - On-demand streaming
 - Content delivery networks (CDNs)



World Wide Web

- **HyperText:** A text in a document which can **hyperlink** to other document.
- **Browsers(Client):** Present materials to the user.
- **Webserver(Server):** Provides access to documents
 - **Uniform Resource Locator (URL):** The address used to identify the documents.
 - **HyperText Transfer Protocol(HTTP):** Transfer the document in the network.

If the server doesn't need to follow the dictionary path to reach the document, these parts can be omitted.



HTML(Hypertext Markup Language)

- Encodes the content of website through **tags**, tells the browser the website's appearance and structure.
- Tags
 - Appearance
 - `<h1> // Heading Level 1(Title)`
 - `<p> // Paragraph`
 - Links to other documents and content
 - `<a href = ...> // Build a hyperlink. href: hypertext reference`
 - Insert images
 - `<img src = ...> // src: source(images's dictionary path or URL)`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My Simple Page</title>
</head>
<body>

  <h1>Hello, HTML World!</h1>

  <p>This is a very simple web page using the basic HTML tags we discussed.
</p>
  <p>Click here to go to
  <a href="https://www.google.com">Google</a>.
  </p>

</body>
</html>
return go(f, seed, [])
}
```

XML(eXtensible Markup Language)

- **Storing and transmitting data**
- **Users can define tags based meaning of data.**
- **A descendant of SGML**

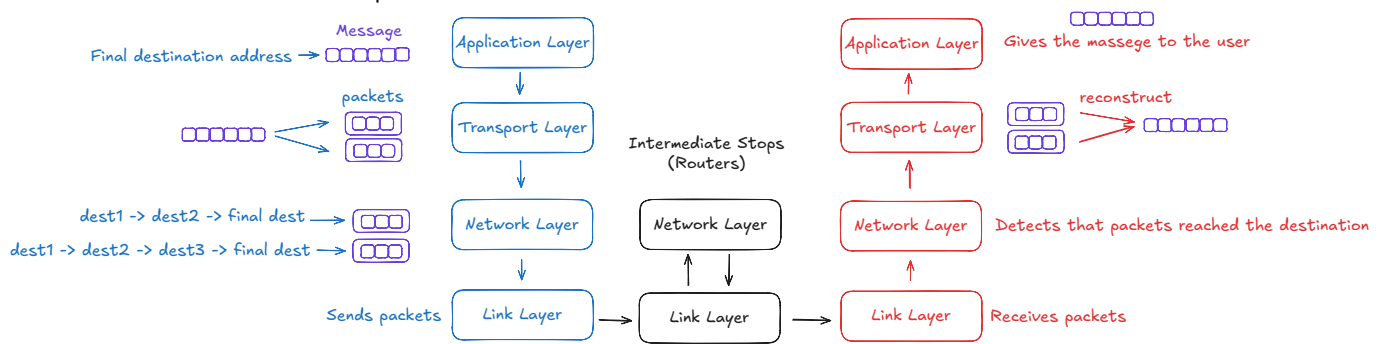
Client-Side and Server-Side Activities

- **Client-Side Activities:** The browser extracts the program and executes it along with HTML to display to users.
 - JavaScript
 - Macromedia Flash
- **Server-side Activities:** Manages data, interacts with databases, and handles client request.
 - Common Gate Way(CGI)
 - Serverlets
 - JavaServer Pages(JSP)/Active Server Pages(ASP)
 - PHP

Internet Software Layers

- Transferring message is through the 4 layers of **software units** in the network.

- **Application:** Constructs the message with address. e.g. Using DNS to turn domain name into numeric IP
- **Transport**
 - Divides long message into small parts to avoid network clogging.
 - Each small part will be assigned a **port number** to help them be reconstructed together at the destination.
 - **Packet:** Consists of small parts, handed to the Network layer.
- **Network:** Uses forwarding table to decide where each packet should go. e.g. Router
- **Link:** Handles actual transmission of packets.



TCP/IP Protocol Suite

- **Transport Layer**
 - Transmission Control Protocol (TCP): Before sending the message, It **establishes the connection**.
 - Establishes the connection: Before sending the message, **will tell the receiver and check it can receive message**.
 - Suitable for unicast(one to one).
 - High reliable but low efficient
 - User Datagram Protocol (UDP): it **just sends** no matter if the receiver is operating.
 - Suitable for multicast, broadcast.
 - High efficient but low reliable
- **Network Layer**
 - Internet Protocol (IP)
 - IPv4 (32 bits)
 - IPv6 (128 bits)

OSI Model

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

Security

- **Attack**
 - **Malware**
 - Self-Replicating & Destructive
 - Virus
 - Worm
 - Trojan Horse
 - Eavesdropping & Information Theft
 - Spyware
 - **Remote Attacks**
 - Denial of Service(DoS): To **send a large number of messages** to a target computer and **make it overload**.
 - Distributed Denial of Service(DDoS)

1. **Install the malware on the lots of computers.**

2. **Command all of those computers** a send large number of messages to target number.

- **Social Engineering & Unsolicited Messages**

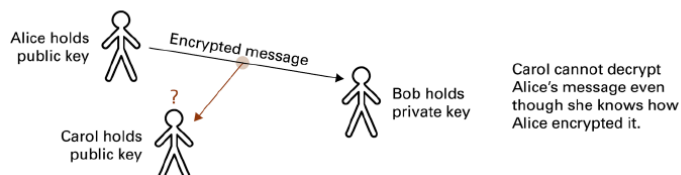
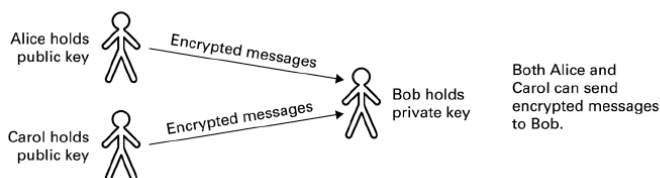
- Phishing
- Spam(can be virus vector)

- **Protection**

- VPN(Virtual Private Network)
- Fire Walls
 - Network-level Protection
 - **Spoofing**^[1] Prevention
 - Individual Computer Protection
 - Specific Protection e.g. Spam filters
- Proxy Server: Acts as a middleman between the client and the target server and can hide client's true identity.
- Auditing software: Monitors activities about network in computer and other connected devices.
- Antivirus software

Encryption

- To convert the message into an **encoded version**.
- **HTTPS**: More secure version of HTTP.
- **TLS(Transport Layer Security)**: Pro version of SSL(Secure Sockets Layer).
- **Public-key Encryption**
 - Public key: Used to encrypt messages
 - Private key: Used to decrypt messages



- **Certificate**: Certificate Authority(CA) verifies public key ownership to establish trust.
- **Digital Signature**: Sender uses private key to encrypt the message.

1. Attackers disguises themselves as a legal entity(like IP) and passes the security mechanisms. ↩